

# Ying Ho, CHOW (Jacky)

Automation Engineer (Robotics)

jackychow.yh@gmail.com +49 (0)160 93416833 Karlsruhe, Germany

https://github.com/jacky-benkyo

## Summary

Results-driven Automation Engineer with 5+ years of experience in system development, integration, and on-site commissioning. Adept at optimizing industrial processes while actively expanding expertise into Modern Robotics and AI. Passionate about leveraging intelligent control systems to drive next-generation manufacturing solutions.

## Skills

- **Programming Languages:** C++, Python, PLC (SCL, LAD)
- **Robotics:** ROS 2 (Humble), MoveIt 2, Gazebo, colcon, CMake, Yaskawa MotoSim EG-VRC
- **Industrial Automation:** TwinCAT 3, TIA Portal, PLC-ANALYZER Pro4
- **Tools & Platforms:** Linux, Docker, Git, GitHub, Jira

## Experience

**SCHMID Group Gebr. SCHMID GmbH** Freudenstadt, Germany

PLC PROGRAMMER – INDUSTRIAL ROBOTICS Jun 2024 – Jun 2026

- Programmed Yaskawa robots for machine loading and unloading using Structured Control Language (SCL) in Siemens TIA Portal.
- Optimized motion sequences of a linear axis unit with a rotary axis, reducing transport time by approximately 19%.
- Independently conducted on-site commissioning for international clients in the USA and Austria.

**Superior Autonomation Ltd** Hong Kong & Shenzhen, China

EMBEDDED SYSTEMS SOFTWARE DEVELOPER (STM32) Mar 2023 – Jul 2023

- Analysed requirements from the Hong Kong Productivity Council (HKPC) and proposed connectivity solutions.
- Tested Wi-Fi 6 modules and conducted on-site integration testing.

**Reactive Robotics GmbH** Munich, Germany

MASTER'S THESIS & SOFTWARE DEVELOPMENT INTERN May 2022 – Jan 2023

- Thesis: *"Conception and Development of a Connectivity Solution for Medical Robot"*.
- Evaluated Over-the-Air (OTA) transmission (4G / Wi-Fi 5 / LTE Cat 1) via MQTT & AMQP using Mosquitto and RabbitMQ.
- Developed a Python-based GUI for remote control features via the TwinCAT ADS communication protocol.
- Applied Test-Driven Development (TDD) principles under Linux; implemented and tested robot control software on Beckhoff PLCs using TwinCAT 3.

**Wandelbots GmbH** Dresden, Germany

WORKING STUDENT – QUALITY ASSURANCE Sep 2020 – Feb 2022

- Increased test coverage for Universal Robots (UR series) by approximately 27%.
- Created automated test cases to improve GUI code coverage and satisfy ISO 10218-1 requirement coverage.

**Nam Wah Intelligent Automation Limited** Hong Kong

AUTOMATION SYSTEMS PROJECT ENGINEER Aug 2014 – Aug 2018

- Executed automation projects across the SDLC, focusing on HMI design and Ladder Diagram (LAD) programming for Allen-Bradley ControlLogix PLCs.
- Authored test scripts and conducted beta testing for Factory Acceptance Testing (FAT) and fire safety inspections.

## Education

**TH Köln (University of Applied Sciences)** MASTER OF ENGINEERING IN AUTOMATION AND INFORMATION TECHNOLOGY • 1,6
Industrial Automation and IT Gummersbach, Germany • Oct 2018 – Jan 2023
Master's thesis completed at Reactive Robotics GmbH.

**Vocational Training Council** DIPLOMA IN ELECTRICAL ENGINEERING
Hong Kong • Sep 2014 – Aug 2017
Part Time

**Hong Kong Baptist University**
BACHELOR OF BUSINESS ADMINISTRATION (HONS.) – MAJOR IN INFORMATION SYSTEMS MANAGEMENT • 2.1
Hong Kong • Sep 2009 – Nov 2011

## Continue Education

**42 Heilbronn** Heilbronn, Germany • Oct 2026 – Present
Project-based computer science curriculum focusing on practical software engineering Admitted after successfully completing the 4-week intensive coding bootcamp ("Piscine"), applying practical fundamentals in C, Linux, and Git. Current focus areas: C/C++, Python, algorithms, data structures, and debugging in a peer-to-peer, peer-reviewed learning environment.

## Side Projects

**Simulation & Control**
Developed and simulated trajectory control for a UR5e robotic arm using ROS 2 (Humble), MoveIt 2, Docker, and RViz, including motion planning and collision checking.

**Embedded Hardware Integration**
Built a low-latency IMU telemetry system (GY-801 sensors) on Raspberry Pi Pico W using micro-ROS.

## Languages

| Cantonese | English | Mandarin | German | Japanese |
|---|---|---|---|---|
| Native | Fluent | Fluent | Conversational (B2) | Basic (A2) |

## Certifications

**German Training Online B2** Apr 2026
Goethe-Institut

**ROS2 Basics in 5 Days (Python)** Apr 2026
The Construct

**JLPT A2 Certificate** Jan 2026
The Japan Foundation

**Zertifikat Deutsch / telc Deutsch B1**
Jun 2024
did deutsch-institut Frankfurt

**Certified Tester Foundation Level (CTFL)**
Jun 2021
ISTQB

## Miscellaneous

**Driving License**
Class B (Hong Kong); conversion to German license in progress (Expected completion: 12/2026)